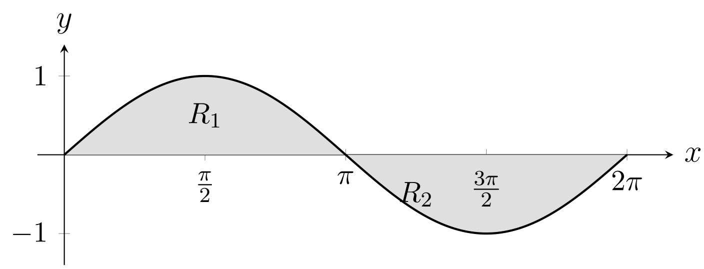
La Scuola d'Italia / Dr. Huson / IB Math: Calculus Name:
1 April 2026

5.11 Classwork: Trigonometric Functions, Identities, & Integration (no calculator)

1. The graph of $f(x) = \sin x$ for $0 \le x \le 2\pi$ is shown. The regions R_1 and R_2 are shaded.



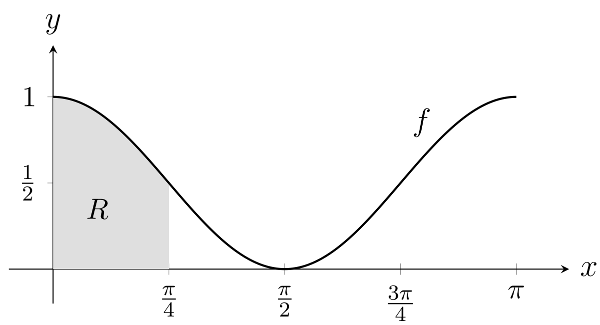
(a) Find $\displaystyle\int_0^{\pi} \sin x \, \mathrm{d}x$. [3]

(b) Write down the value of $\displaystyle\int_{\pi}^{2\pi} \sin x \, \mathrm{d}x$. [1]

(c) Explain why $\displaystyle\int_0^{2\pi} \sin x \, \mathrm{d}x = 0$, even though the shaded regions clearly have area. [2]

(d) Find the total area enclosed between the graph of f and the x-axis from $x = 0$ to $x = 2\pi$. [1]

2. The graph of $f(x) = \cos^2 x$ for $0 \leq x \leq \pi$ is shown. The shaded region R is bounded by the graph and the x-axis from $x = 0$ to $x = \dfrac{\pi}{4}$.

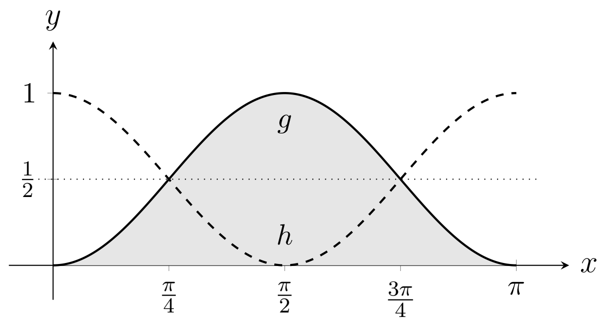


(a) Using the identity $\cos 2x = 2\cos^2 x - 1$, show that $\cos^2 x = \dfrac{1}{2}(1 + \cos 2x)$. [2]

(b) Hence find $\displaystyle\int \cos^2 x \, \mathrm{d}x$. [3]

(c) Find the exact area of the shaded region R. [3]

3. The graphs of $g(x) = \sin^2 x$ (solid) and $h(x) = \cos^2 x$ (dashed) are shown for $0 \le x \le \pi$. The dotted line is $y = \frac{1}{2}$. The shaded region is bounded by g and the x-axis.



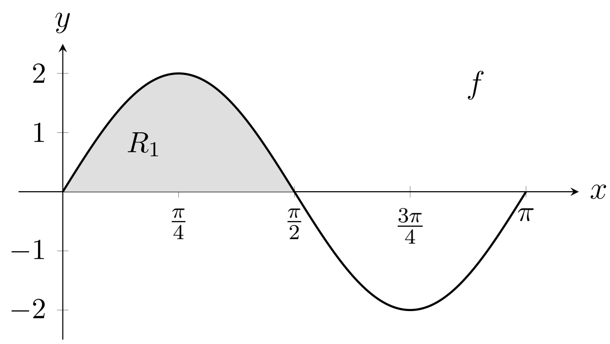
(a) The graphs intersect at $x = \dfrac{\pi}{4}$. Verify this by evaluating $\sin^2\dfrac{\pi}{4}$ and $\cos^2\dfrac{\pi}{4}$. [2]

(b) Show that $\sin^2 x = \dfrac{1}{2}(1 - \cos 2x)$. [2]

(c) Find $\displaystyle\int_0^{\pi} \sin^2 x \,\mathrm{d}x$. [4]

(d) Without further integration, find $\displaystyle\int_0^{\pi} \cos^2 x \,\mathrm{d}x$. Justify your answer. [2]

4. The function f is defined by $f(x) = 4\sin x \cos x$ for $0 \le x \le \pi$. The graph of f is shown. The shaded region R_1 is bounded by the graph and the x-axis from $x = 0$ to $x = \dfrac{\pi}{2}$.



(a) Using the identity $\sin 2x = 2\sin x \cos x$, show that $f(x) = 2\sin 2x$. [2]

(b) Write down the period of f. [1]

(c) Find the exact area of R_1. [4]

(d) Find the total area enclosed between the graph of f and the x-axis from $x = 0$ to $x = \pi$. Give a reason. [2]

(e) Find $f'(x)$ and hence find the x-coordinate where f has its maximum value on $\left[0, \dfrac{\pi}{2}\right]$. [2]